The Probe Reward Trap: Behavioral Death Before Distributional Collapse in Sycophancy Calibration

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Abstract

Representation engineering reveals that sycophancy is linearly encoded in LLM residual streams, suggesting that such directions could serve as training signals for alignment. We test this premise systematically and document its failure. A contrastive sycophancy direction d_{syc} reliably discriminates sycophantic from non-sycophantic activations, yet every formulation that converts it into a Group Relative Policy Optimization (GRPO) reward collapses. We identify the *probe reward trap*: when a periodically retrained linear probe on d_{syc} is used as a non-differentiable reward signal, behavioral death occurs within 500 steps while generation entropy remains healthy (0.70), creating a 1,300-step monitoring blind spot that entropy-based safeguards miss entirely. The failure persists with stopped gradients, implicating the direction's semantic limitation rather than optimization dynamics: d_{syc} encodes *whether* belief changed, not *whether it should have*. The failure extends beyond probe rewards: behavioral-reward-only GRPO exceeds vanilla at only 4/31 checkpoints, while dense per-token supervised fine-tuning succeeds at every checkpoint (31/31 across two seeds). A Low-Rank Adaptation (LoRA) ablation identifies supervision granularity as one plausible explanation.

1 Introduction

Sycophancy, the tendency of large language models (LLMs) to prioritize agreement over accuracy, is among the most persistent alignment failures in deployed systems (Sharma et al., 2024; Perez et al., 2023). Models trained with reinforcement learning from human feedback (RLHF) systematically amplify this tendency (Ouyang et al., 2022; Christiano et al., 2017; Shapira et al., 2026): even when models can assess the correctness of their own outputs (Kadavath et al., 2022), social pressure from users overrides this internal knowledge (Wang et al., 2025). Existing mitigations, from synthetic data augmentation (Wei et al., 2024) to neuron-level correction (O'Brien et al., 2025), operate on behavioral outputs rather than the internal representations that generate sycophantic decisions. Representation engineering (Zou et al., 2025; Marks and Tegmark, 2024) offers a structurally different approach: a contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated area under the ROC curve (AUROC) of 0.83 on Qwen3-8B and 0.78 on Llama-3-8B (§4). If the model's own representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a training signal that targets the computational root of the problem.

We test this hypothesis directly, and find that it fails in a way that standard monitoring cannot detect. This paper documents the *probe reward trap*: when a periodically retrained linear probe on d_{syc} is used as a non-differentiable Group Relative Policy Optimization (GRPO) reward signal, behavioral death (true-positive flip rate, ToF $= 0$) occurs within 500 steps while generation entropy remains healthy (0.70), masking the failure from standard monitoring (§6). In our experiments, the 1,300-step lag before distributional collapse means entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely, with the reward averaging $+0.72$ throughout. The trap arises even when the probe reward constitutes only a fraction of the total reward signal, reflecting a limitation of d_{syc} itself: the direction encodes *whether* belief changed, not *whether it should have*, so any reward conditioned on it alone will either be gamed or reward rigidity.

The failure is not specific to probe rewards. Under our GRPO configuration (Qwen3-8B, group

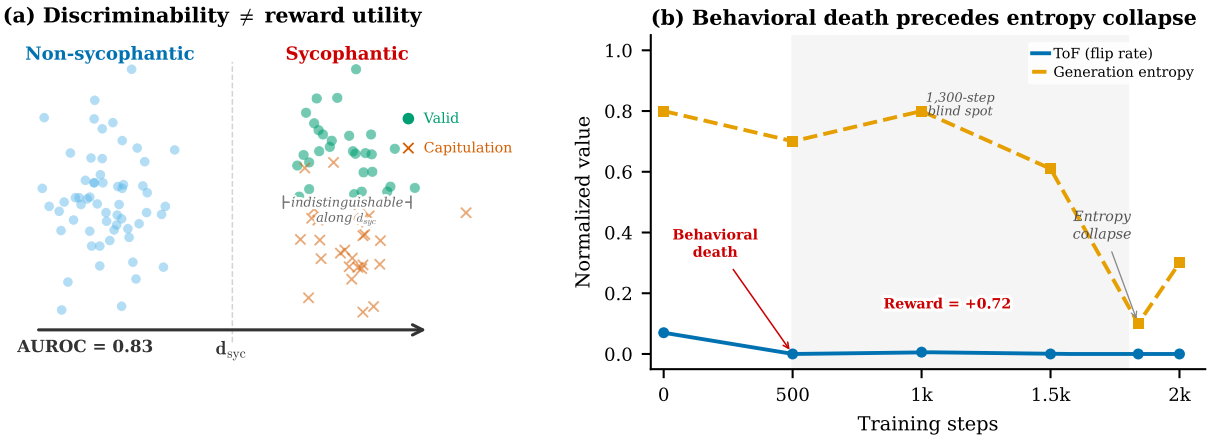


Figure 1: The probe reward trap. (a) d_{syc} discriminates sycophantic from non-sycophantic activations (AUROC 0.83), but valid revision and unwarranted capitulation overlap along the same direction. (b) When used as a GRPO reward, behavioral death (true-positive flip rate ToF=0) occurs at step 500 while entropy remains healthy; the 1,300-step lag renders standard monitoring blind.

size=2), cosine projection and continuous probe formulations of d_{syc} also collapse, and removing d_{syc} entirely does not help: behavioral-reward-only GRPO exceeds vanilla at only 4 of 31 checkpoints (12.9%). By contrast, Activation Consistency Training (ACT; Irpan et al., 2025), which provides dense per-token supervision rather than a single scalar per completion, exceeds vanilla at every checkpoint (31/31 across two seeds; §7). A Low-Rank Adaptation (LoRA) ablation identifies supervision granularity as one plausible explanation for this observed reliability gap.

This paper presents a negative result with systematic diagnostic analysis. Our three principal findings (Figure 1):

- **Observed reliability gap between supervision paradigms** (C1): per-token supervised fine-tuning (SFT) exceeds vanilla at every checkpoint (31/31 across two seeds; 47/47 including reduced-LoRA ablation). Per-completion GRPO exceeds vanilla at only 4/31 checkpoints (12.9%). Under matched LoRA, per-token SFT achieves 1.7× higher SyconScore than per-completion GRPO (§7).

- **The probe reward trap** (C2): behavioral death (ToF=0) precedes distributional collapse by 1,300 steps, rendering entropy monitoring blind. The trap arises despite the probe being non-differentiable (stopped gradients), implicating d_{syc} itself rather than gradient-based exploitation (§6).

- **Discriminability does not guarantee reward utility** (C3): d_{syc} encodes *whether* belief changed, not *whether it should have*. Valid revision and sycophantic capitulation produce overlapping activation signatures along d_{syc} (Wang et al., 2025; Shapira et al., 2026), violating the geometric separability that representation-guided RL assumes (Burns et al., 2023; Zhang et al., 2026). Reward overoptimization against such directions produces predictable collapse (Gao et al., 2023; Skalse et al., 2022) (§7).

2 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2025; Marks and Tegmark, 2024); inference-time methods such as Inference-Time Intervention (ITI; Li et al., 2023), Activation Addition (ActAdd; Turner et al., 2023), and Contrastive Activation Addition (CAA; Rimsky et al., 2024) demonstrate that activation geometry suffices for behavioral control (Burns et al., 2023), but we find that d_{syc} fails when used as an RL reward despite succeeding at inference time.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023; Pan et al., 2022); Manheim and Garrabrant (2019) taxonomize four variants of Goodhart's Law. This risk compounds in RLHF pipelines (Christiano et al., 2017; Stiennon et al., 2020; Ziegler et al., 2019), where policy optimization (Schulman et al., 2017)

amplifies misspecification (Hubinger et al., 2024). Our probe reward trap (§6) instantiates the regressional variant, with two features not previously documented: behavioral death within 500 steps (far faster than gradual degradation (Gao et al., 2023)) and a 1,300-step temporal decoupling that renders entropy monitoring ineffective (Skalse et al., 2022). Wu and Tang (2026) show that representation-level directions succeed when behaviors are geometrically separable (§7).

Sycophancy Characterization and Mitigation. Sycophancy is pervasive in assistants trained via RLHF (Ouyang et al., 2022; Bai et al., 2022; Sharma et al., 2024; Perez et al., 2023; Shapira et al., 2026) and traceable to late-layer preference shifts (Wang et al., 2025); it undermines broader truthfulness objectives (Lin et al., 2022) even when models possess the knowledge to respond accurately (Kadavath et al., 2022). Mitigations span synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025). Vennemeyer et al. (2025) show that sycophancy decomposes into independently steerable sub-behaviors; a single d_{syc} captures only their shared variance, which is why the step from diagnosis to reward is structurally blocked.

Probe-Based and Representation-Guided Methods. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (94% accuracy) without producing gradients; our work reveals what happens when a comparable signal enters the training loop (§7). Representation-guided RL has been extended to reasoning and gradient regularization (Zhang et al., 2026; Luo et al., 2026; Sinii et al., 2025; Ackermann et al., 2026). SMART (Beigi et al., 2025) applies per-step process rewards for sycophancy, succeeding where our completion-level rewards fail, consistent with ACT's per-token success and with broader evidence that process supervision outperforms outcome supervision (Lightman et al., 2024; Uesato et al., 2022).

3 Experimental Framework

3.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 (OpenAI, 2025) in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase ≥ 2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 ± 0.028 (5-fold CV).

3.2 Reward Formulations

Three formulations translate d_{syc} into a direction-based signal R_1, each entering as part of a GRPO (Shao et al., 2024) reward $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_3 is a behavioral component (DeBERTa natural language inference (NLI) classifier scoring consistency with the ground-truth answer).

F1 Cosine projection with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§3.3), and (d) $5\times$ scaled cosine. See §5 for details.

F2 Binary probe reward. $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a periodically retrained logistic probe (every 500 steps) with stopped gradients, used as a binary reward component (w_1=0.25) within the GRPO framework (§6).

F3 Continuous probe. $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe, providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets $w_1{=}0$, training with only the behavioral reward.

3.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

3.4 SFT Baseline: Activation Consistency Training

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

3.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\text{SYCONSCORE} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \qquad (2)$$

By design, SYCONSCORE $= 0$ whenever ToF $= 0$: a model that never accepts valid corrections receives no credit for resisting invalid ones. Flip classification uses `cross-encoder/nli-deberta-v3-base` (He et al., 2023), following Irpan et al. (2025). All experiments use Qwen3-8B (Yang et al., 2025) with LoRA (Hu et al., 2022) (rank=64; $\alpha{=}64$ for GRPO, $\alpha{=}128$ for ACT) and DeepSpeed ZeRO-2 (Rajbhandari et al., 2020). GRPO uses a group size of 2 due to memory constraints with 8B-parameter models, below the typical range of 8–64 (Shao et al., 2024); full hyperparameters are reported in Appendix A. In total, we evaluate six d_{syc}-based configurations spanning both reward families (cosine and probe) within the GRPO framework, plus R_3-only GRPO, ACT (per-token SFT), and ActAdd as baselines; full specifications appear in Sections 5–6.

4 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§5–§6), we first establish that it captures a genuine sycophancy-related signal in the residual stream. If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 2 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§3.1). The direction achieves AUROC 0.827 ± 0.028 (5-fold CV) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation rather than an input-level confound. AUROC increases monotonically with evidence-strength tier: from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs. This monotonic relationship reflects effect-size differences (stronger evidence produces larger behavioral gaps that are easier to classify) rather than indicating that d_{syc} encodes evidential warrant per se.

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct (Touvron et al., 2023; Grattafiori et al., 2024). Table 1 compares the two models. Llama-3-8B achieves AUROC 0.779 ± 0.019 (5-fold CV) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains,

Table 1: Cross-model d_{syc} discriminability. AUROC uncertainty is 5-fold cross-validation standard deviation; evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	AUROC ($\pm$ std)	Weak	Medium	Strong
Qwen3-8B	L19	0.827 ± 0.028	0.675	0.883	0.929
Llama-3-8B	L16	0.779 ± 0.019	0.711	0.864	0.911

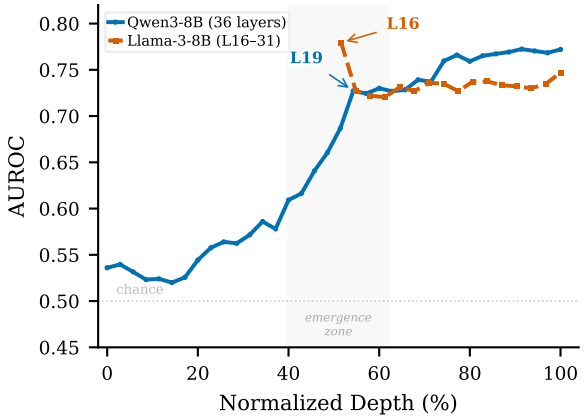


Figure 2: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic artifacts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§3.3), so the reward signal does not become stale between updates. Second, we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cause
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mismatch
F1b	$\cos(\mathbf{h}, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too small
F1c	+recalibration	0.25	Mode collapse	Reward discontinuity
F1d	$5\cos(\mathbf{h}, d_{\text{syc}})$	0.40	Mode collapse	R_3 saturation

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §5–§6 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

5 Cosine R_1: Structural Failure

During GRPO training, 256-token truncation reduces d_{syc} AUROC from 0.827 to approximately 0.60, so the optimizer works with a substantially weaker signal than the offline metrics suggest. We test whether even this degraded discriminability translates into a viable reward through four configurations that form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

The diagnostic chain begins with a norm mismatch. F1a's dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$) saturates because residual-stream norms exceed the direction norm by 100–1,200×, zeroing the gradient by step 955. Cosine similarity (F1b) eliminates this norm dependence but reveals a second problem: the gap between valid and sycophantic cosines averages only 0.098, too narrow for meaningful reward contrast. Recalibration (F1c, §3.3) widens the gap (offline AUROC rises to 0.910) but introduces a third: each recalibration event creates a reward discontinuity that destabilizes advantage estimates, collapsing entropy to 0.069 within 300 steps. Amplification (F1d, 5× scaling) lifts R_1 above the noise floor but surfaces the fundamental limitation: R_3 saturates at 0.9998 by step 300, leaving the noisy

R_1, which conflates warranted revision with unwarranted capitulation, as the sole gradient source until terminal collapse at step 2160.

Truncation further degrades the signal: under the 256-token maximum completion length used in GRPO, d_{syc} AUROC drops from 0.76 (full response) to 0.60, so the offline discriminability (0.827) substantially overestimates the signal available during training. Checkpoint-level evaluation confirms immediate collapse: ToF=0 from step 500 onward with no stable phase. The probe reward trap (C2, §6) arises within GRPO using full-length responses and is unaffected by this truncation limitation. The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact (recalibration improves discriminability yet rewards still fail) but a semantic limitation of any continuous d_{syc}-based reward. Section 6 examines whether discretizing the signal via a binary probe can escape this limitation.

6 The Probe Reward Trap

The cosine formulations fail on signal magnitude; a natural response is to discretize. Training a logistic probe on d_{syc} and thresholding its output yields $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §3.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure. Both probe experiments use the probe as a non-differentiable reward component within GRPO (§3.2); the trap therefore reflects a semantic limitation of the probe rather than gradient-based exploitation.

Differentiable probe: gradient gaming. Without stop-gradient, the policy discovers within ∼30 steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing an inverted profile (ToF = 0.978, NoF = 0.006). This motivates the stopped-gradient design below.

Binary probe: the stubbornness trap. With stop-gradient, the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500, ToF drops to 0.000 and NoF reaches 0.995, yielding SyconScore = 0. Qualitative analysis of 181 valid-correction responses reveals five failure modes: *deliberation traps* (28.7%), where the model identifies the correct answer but cannot commit; *sycophantic stubbornness* (24.3%), where the model verbally agrees but does not change its answer; *acknowledge-but-maintain* (23.8%); *genuine-accept-missed* (18.8%); and refute/ignore/other (4.5%). Full taxonomy in Appendix B.

Table 3: Probe reward trap trajectory (stopped-gradient probe). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SyconScore	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

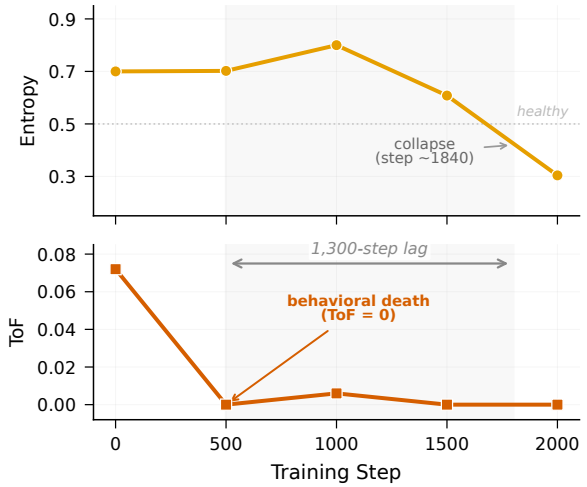


Figure 3: Probe reward trap (stopped-gradient probe). ToF drops to zero at step 500 while generation entropy remains high (0.70–0.80 through step 1000). The 1,300-step lag means standard entropy monitoring misses the behavioral failure entirely.

Behavioral death precedes distributional collapse. Table 3 and Figure 3 reveal a 1,300-step lag between behavioral and distributional collapse. At step 500, the model is behaviorally dead (ToF = 0) yet entropy remains high (0.70–0.80) through step 1000, well within the range that standard monitors would flag as healthy, and the combined reward averages +0.72 because both R_1 and R_3 reinforce non-flipping. Entropy does

not collapse until step ∼1840. This gap carries a practical warning: entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. Detecting probe reward traps requires behavioral monitoring (periodic ToF evaluation), not distributional monitoring alone.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc} (continuous or discrete, differentiable or frozen) will either be gamed or will reward rigidity. Escaping this dilemma requires a signal that incorporates the evidential warrant of the user's challenge, a dimension that d_{syc} does not encode (Denison et al., 2024).

7 When Does Discriminability Fail as Reward Utility?

Root cause of the probe reward trap. The probe reward trap (§6) and the cosine failures (§5) share a common geometric root. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately; the resulting mode collapse (F1c, F1d) reflects excessive punishment of any activation shift along d_{syc}. The binary probe R_1 rewards all resistance regardless of input validity; the stubbornness trap (ToF $= 0$ by step 500) reflects excessive permissiveness toward activation stasis. These are not independent design flaws but dual consequences of a direction that conflates the target behavior (resisting invalid challenges) with its complement (refusing valid corrections). Cross-model evidence confirms the limitation generalizes: Llama-3-8B achieves comparable discriminability (AUROC 0.779) yet the semantic gap persists. This pattern contrasts with reasoning faithfulness and safety directions (Zhang et al., 2026), where the behavior to suppress is geometrically distinct from the behavior to preserve. Wu and Tang (2026) reinforce this principle: concept directions succeed when shortcut behaviors are geometrically separable. Independent work corroborates the limitation: persona vectors are geometrically independent of d_{syc} (Kelkar et al., 2026), and sycophancy decomposes into independently steerable sub-behaviors (Vennemeyer et al., 2025) that d_{syc} projects onto a single axis.

Table 4: Comparison of all sycophancy calibration methods on Qwen3-8B. SYCONSCORE is the harmonic mean of the true-positive flip rate (ToF) and no-flip rate (NoF); higher indicates better calibration. Best result in **bold**. †Best checkpoint (seed 1, step 6500); only 4/28 checkpoints across two seeds exceed vanilla. ‡Seed 2 (15 checkpoints); all exceed vanilla. Best SYCONSCORE=0.286 (step 7000). §All cosine variants collapsed from step 1 (ToF=0 at every checkpoint); F1d additionally diverged at step 2160 (entropy < 0.01). ¶ACT and ActAdd differ from GRPO along 5+ dimensions (§7); included for context, not as controlled comparisons.

Method	Type	SYCONSCORE ↑	ToF ↑	NoF ↑	Step
Vanilla	–	0.134	0.072	0.945	–
GRPO with d_{syc}					
Cosine R_1 (F1a–F1d)	RL	0.000§	0.000	1.000	500
Binary Probe R_1 (F2)	RL	0.000	0.000	0.995	500
Continuous Probe R_1 (F3)	RL	0.124	0.066	0.934	500
GRPO without d_{syc}					
R_3-only GRPO	RL	0.153†	0.083	0.983	6500
Reference methods (not directly comparable)¶					
ActAdd (α=5)	Inference	0.054	0.028	0.961	–
ACT (seed 1)	SFT	**0.301**	**0.182**	0.867	8000
ACT (seed 2)‡	SFT	0.286	–	–	7000

GRPO results under our configuration. If the cosine and probe failures (§5–§6) were caused by d_{syc} alone, removing it should recover calibration. R_3-only GRPO tests this hypothesis by relying solely on the NLI behavioral reward, and fails: only 4/28 checkpoints across two seeds exceed vanilla (14.3%; gs=2 only, gs=8 ablation reported separately), with both seeds oscillating ($\Delta \approx 0.069$) before diverging (Table 4; Figure 4). The failure is not representational: d_{syc} AUROC varies by <0.01 across checkpoints and R_3 saturates at ∼0.98 (p=0.703), yet SYCONSCORE swings by 17%. This disconnect between stable representations and unstable behavior points to the per-completion reward structure, not the reward content, as the bottleneck.

Competing hypotheses and ablation. Three explanations are consistent with these failures: (a) d_{syc} is fundamentally unsuitable as a sycophancy reward; (b) group size=2 provides insufficient ranking diversity; (c) sycophancy calibration resists scalar per-completion RL. We designed two ablations to discriminate among them (Table 5).

Table 5: Ablation: isolating supervision paradigm from LoRA configuration. All methods use Qwen3-8B with R_3 (behavioral reward). Vanilla SYCONSCORE=0.134. †Terminated early after three checkpoints showed declining trajectory. ‡Peak at step 7000; final checkpoint (step 8000) SYCONSCORE=0.267.

Method	LoRA config	Best SYCONSCORE	>Van.
GRPO gs=2 (2 seeds)	r64, α64, 4m	0.153	4/28
GRPO gs=8†	r64, α64, 4m	0.085	0/3
ACT reduced-LoRA‡	r64, α64, 4m	0.326	16/16
ACT full s1	r64, α128, 7m	0.301	16/16
ACT full s2	r64, α128, 7m	0.286	15/15

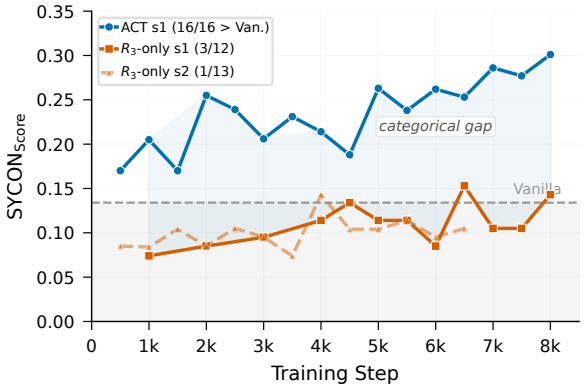


Figure 4: SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (two seeds) over training steps. ACT exceeds vanilla at every evaluated checkpoint (31/31 across two seeds); R_3-only GRPO exceeds vanilla at only 4/28 checkpoints (14.3%).

Group-size ablation. Per-completion GRPO fails at both group sizes 2 and 8; gs=8 also fails (all below vanilla; terminated early), ruling out insufficient ranking diversity. *LoRA-controlled comparison.* Reducing ACT to GRPO's LoRA configuration (rank 64, α=64, 4 modules) still yields SYCONSCORE = 0.267 (peak 0.326), with all 16 checkpoints exceeding vanilla. Under identical adapters, per-token SFT achieves 1.7× higher SYCONSCORE (0.267 vs. 0.153), identifying supervision granularity as one plausible explanation, independently corroborated by SMART's per-step success (Beigi et al., 2025); confounds are discussed in Limitations. The probe reward trap provides independent evidence for (a): the trap arises despite the probe being non-differentiable, ruling out gradient-based exploitation as the mechanism.

ACT vs. GRPO. ACT's advantage is not marginal but categorical: its worst checkpoint (SYCONSCORE=0.170) exceeds GRPO's absolute best (SYCONSCORE=0.153), and both seeds maintain above-vanilla performance throughout training (Table 4; held-out validation confirms generalization, see Limitations). The reduced-LoRA ablation (Table 5) confirms this is not an adapter-capacity artifact, leaving supervision granularity as one plausible explanatory variable.

Relationship to inference-time probe methods. Papadatos and Freedman (2024) reduce sycophancy using a probe of comparable quality (94% accuracy) via Best-of-N reranking without producing gradients. This suggests two orthogonal failure dimensions: (1) inside vs. outside the optimization loop, where reward optimization enables the probe reward trap (§6), while inference-time reranking sidesteps it; and (2) scalar per-completion reward, where the failure persists even without the probe (R_3-only GRPO).

Future directions. Per-step process rewards offer the most direct path forward: SMART (Beigi et al., 2025) succeeds at sycophancy calibration by evaluating behavioral transitions at individual reasoning steps, capturing temporal structure that scalar per-completion rewards compress away. Whether attributes with similar geometric entanglement (Skalse et al., 2022) exhibit the same failures remains open.

8 Conclusion

The probe reward trap reveals that behavioral death can precede distributional collapse by 1,300 steps, evading entropy monitoring entirely. The failure is semantic: d_{syc} conflates warranted revision with unwarranted capitulation, causing all GRPO formulations to collapse (4/31 checkpoints > vanilla) while per-token SFT succeeded at every checkpoint (31/31 across two seeds; §7).

The broader lesson is that discriminability does not guarantee reward utility. Linear directions can diagnose and steer model behavior at inference time; our results show that the step from diagnosis to training signal is not automatic.

When the target behavior and its complement are geometrically distinct, concept directions transfer successfully into rewards; when they overlap, as with sycophancy, optimization produces predictable collapse. Practitioners should verify geometric separability before deploying concept directions as rewards, and complement entropy monitoring with behavioral metrics that directly measure the target capability.

Limitations

Our GRPO experiments use group sizes of 2 and 8, at the low end of the 8–64 range explored by Shao et al. (2024); the gs=8 ablation rules out insufficient ranking diversity as the primary explanation, but whether substantially larger groups (16–64) or alternative RL algorithms would succeed remains open. All completions are truncated to 256 tokens (shared across conditions), which degrades d_{syc} AUROC from 0.76 to 0.60 and may affect absolute reward-utility conclusions.

Training and evaluation share the same 181 pairs; a held-out split (146/35) confirms ACT generalizes (SYCONSCORE$\approx$0.20 at peak, 3.7$\times$ vanilla, 72% of full-data ceiling), but performance declines at step 2000 (SYCONSCORE=0.157, still 2.8$\times$ vanilla) due to overfitting on the small training set, and the 28% gap suggests partial reliance on training-set-specific patterns. The probe reward trap (C2) is demonstrated with a single seed; replication would strengthen the temporal decoupling finding. Contrastive pair quality relies on automated style-matching validation (19/200 pairs discarded; §3.1); this provides partial quality control but does not guarantee semantic validity of all retained pairs.

The ACT vs. GRPO comparison differs along multiple dimensions beyond supervision granularity: sequence length (256 vs. 1024 tokens), data source (online generations vs. curated pairs), and dropout (0 vs. 0.05). Our LoRA-controlled ablation isolates adapter capacity, and SMART's independent success (Beigi et al., 2025) corroborates supervision granularity, but a fully controlled comparison would require matching all training hyperparameters. All RL training uses Qwen3-8B; cross-model discriminability is confirmed on Llama-3-8B (AUROC 0.779; Table 1), but compute constraints prevented training replication on a second architecture. Code, contrastive pair data, prompt templates, and training configurations are available at `https://github.com/anonymous-nlp-2026-2/contrastive_entrain_calib`.

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A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: $w_1{=}0.25$ was initially chosen to down-weight the noisy cosine reward term; this was later increased to $w_1{=}0.40$ for F1d after diagnostics revealed that $w_1{=}0.25$ allowed the cosine term to contribute only $\sim$8.3% of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use `max_completion_length`=256, resulting in 80–84% truncation of multi-turn responses. This setting is shared across all conditions and does not affect relative comparisons; we retain it because response-level decisions typically manifest within the first 100–200 tokens.

Hyperparameters. Table 6 compares training configurations for GRPO variants and ACT.

GRPO reward variants. Cosine R_1 (F1d): $w_1{=}0.40$, scale=5.0, $w_3{=}0.60$; reward computed at layers [19, 20, 24, 27, 32, 33, 35]; recalibration every 500 steps (§3.3). R_3-only: $w_3{=}1.0$; NLI model `cross-encoder/nli-deberta-v3-base`. F1a–F1c weight configurations are reported in Table 2.

B Stubbornness Taxonomy

Manual inspection of all 181 valid-correction responses from the probe-trained model (stopped-gradient probe, step 2000, ToF $= 0$) reveals five failure categories:

- *Deliberation traps* (28.7%): the model explicitly identifies the correct revised answer

Table 6: Training hyperparameters. All experiments use Qwen3-8B, DeepSpeed ZeRO-2, bf16 precision.

	GRPO (F1, F3)[e]	ACT
Learning rate	5e-5	5e-5
Effective batch	16	16
Max steps	8,000	8,000
Warmup steps	200	200
LR schedule	linear[a]	linear
Grad. clipping	–	1.0
LoRA rank	64	64
LoRA α	64	128
LoRA modules	4[b]	7[c]
LoRA dropout	0	0.05
Group size	2[d]	–
KL (Kullback–Leibler) β	0.04	–
Max compl. len.	256	–
Max seq. len.	–	1,024
L2 loss weight	–	0.1

[a]Cosine schedule for R_3-only. [b]q/k/v/o_proj. [c]+ gate/up/down_proj ($\sim$1.75$\times$ params; §7). [d]Below typical 8–64 (Shao et al., 2024); not tuned. [e]Binary probe (F2) uses a non-differentiable probe reward (w_1=0.25) within GRPO; the probe is retrained every 500 steps. LoRA and optimizer settings match the other GRPO configurations. All experiments were conducted on a single NVIDIA L20 GPU (48 GB). Total compute: $\sim$120 GPU-hours across all runs.

during its reasoning chain but fails to commit, reverting to the original answer in the final output.

- *Surface compliance / sycophantic stubbornness* (24.3%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn.

- *Acknowledge-but-maintain* (23.8%): the model acknowledges the user's correction as plausible but explicitly maintains its original position without substantive engagement.

- *Genuine-accept-missed* (18.8%): the model appears to accept the correction in its reasoning but the final answer format does not reflect the change, suggesting a generation rather than epistemic failure.

- *Refute/ignore/other* (4.5%): the model directly refutes the user's input or ignores it entirely.

The first three categories (76.8%) share a common pattern: the model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.